Reaction Networks Analysis and Mechanistic Catalytic Cycles for the Enzymatic Depolymerization of PET

Igor Gamm[1], Tobias Heinks[2], Katrin Hofmann[3], Simon Last[2], Luise Blach[1], Ren Wei[4], Uwe T. Bornscheuer[4], Jan von Langermann[2] and Christof Hamel[1]

[1]Institute of Process Engineering and [2]Institute of Chemistry at Otto von Guericke University Magdeburg/Germany; [3]Anhalt University of Applied Sciences, Process Engineering Koethen/Germany; [4]University of Greifswald, Institute of Biochemistry

Introduction

PET (polyethylene terephthalate) is one of the most used plastics worldwide, primarily utilized in fibers, films, and containers. Since virgin PET is inexpensive to produce and mechanically recycled PET often exhibits significantly different properties [1-3], recycling is still now economically unattractive in many countries [4]. As a result, a large portion of PET is either incinerated or ends up in the ocean, posing a significant environmental burden [5]. Enzymatic depolymerization is a very promising process that offers mild reaction conditions, environmental friendliness, improved energy efficiency and new economic opportunities [4]. In the literature, enzymatic depolymerization is typically studied with focus on the total depolymerization production of terephthalic acid (TPA) as a precursor for PET re-synthesis [4]. However, other potential products for re-synthesis with increasing demand are Mono-(2-hydroxyethyl) terephthalate (MHET) and Bis-(2-hydroxyethyl) terephthalate (BHET). This study provides a deeper insight into the underlying mechanistic pathways and reactions involved in their formation [6].

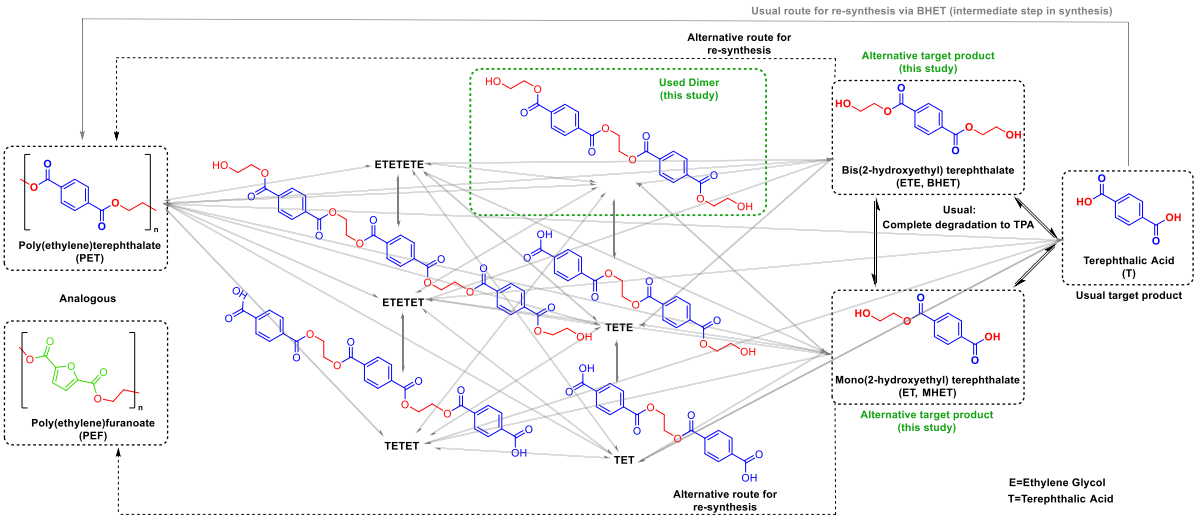

Figure 1: Network of the enzymatic depolymerization of PET including functional intermediates

Development of Mechanistic Kinetics

Based on a promising preselected enzyme, identified in a previous screening of 12 ones [6], kinetic analysis and reduction of the reaction (sub-)network as illustrated in

Fig. 1 is conducted. Own experimental findings and literature suggest that different enzymes have varying preferences for binding sites [7], and various potential mechanistic cycles for the soluble dimer intermediate (ETETE) could be derived. The dimer provides initial insights into the preferred choices of enzyme binding sites, as schematically shown in Fig. 2. These cycles are intended to be used in mechanistic kinetic model development of different complexity for process design and optimization.

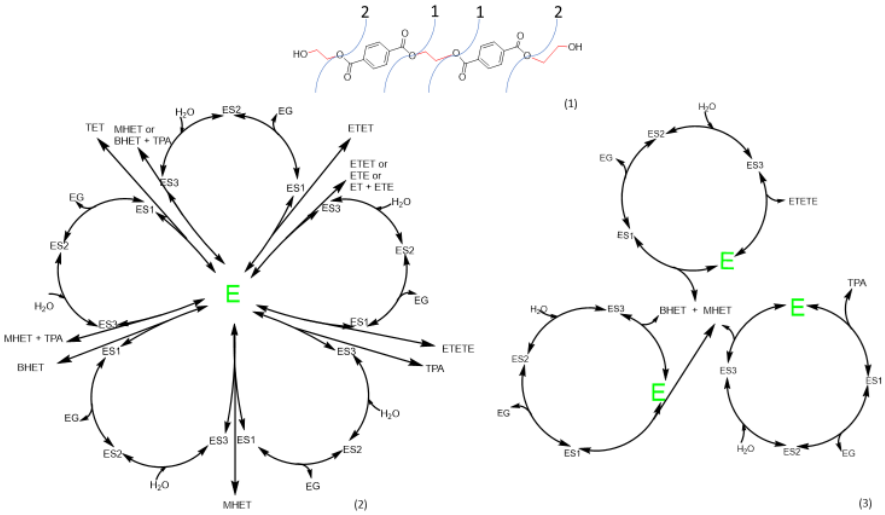

Figure 2: (1) Postulated enzyme binding sites based on the dimer, (2) potential mechanistic cycle assuming random selection of enzyme binding sites, (3) potential mechanistic cycle assuming a preferred central enzyme binding site.

Results and Outlook

A reduced reaction network for enzymatic PET depolymerization is identified based on reliable HPLC analysis of all species involved. Three potential reaction pathways are identified and mechanistic cycles are derived, representing random binding, central binding site preference, and outer binding site preference as illustrated in Fig. 2. The models aim to enable future data-driven comparisons, providing deeper insights into enzyme-specific depolymerization mechanisms. In the next steps, the mechanistic models will be extended to incorporate parallel non-enzymatic depolymerization, competitive inhibition, and application to the trimeric oligomer ETETETE.

[1] a) S. L. Wong, N. Ngadi, T. Abdullah, I. M. Inuwa, *Renew. Sustain. Energy Rev.* **2015**, *50*, 1167–1180; b) R. O. Ebewele, *Polymer science and technology*, CRC Press, Boca Raton, **2000**; c) D. V. Rosato, D. P. Di Mattia, D. V. Rosato in *Designing with plastics and composites*. *A Handbook* (Eds.: D. V. Rosato, D. P. Di Mattia, D. V. Rosato), Springer Verlag, Boston, MA, **2014**; d) L. Desidery, M. Lanotte in *Plastic waste for sustainable asphalt roads* (Eds.: F. Giustozzi, S. Nizamuddin), Elsevier, Amsterdam, **2022**;

[2] A. B. Raheem, Z. Z. Noor, A. Hassan, M. K. Abd Hamid, S. A. Samsudin, A. H. Sabeen, *J. Clean. Prod.* **2019**, *225*, 1052–1064.

[3] A. Majumdar, S. Shukla, A. A. Singh, S. Arora, *Resources, Conservation & Recycling* **2020**, 161, 104915

[4] V. Tournier, S. Duquesne, F. Guillamot, H. Cramail, D. Taton, A. Marty, I. André, *Chem. Rev.* **2023**, *123*, 5612–5701.

[5] a) J. R. Jambeck, R. Geyer, C. Wilcox, T. R. Siegler, M. Perryman, A. Andrady, R. Narayan, K. L. Law, *Science* **2015**, *347*, 768–771; b) X. Zhu, C. M. Rochman, B. D. Hardesty, C. Wilcox, *Deep-Sea Res. I: Oceanogr. Res. Pap.* **2024**, *206*, 104266; c) R. Geyer, J. R. Jambeck, K. L. Law, *Sci. Adv.* **2017**, *3*, e1700782;

[6] T. Heinks, K. Hofmann, S. Last, I. Gamm, I. Blach, R. Wie, U. T. Bornscheuer, C. Hamel, J. v. Langermann, *ChemSusChem* **2024,** e202401759

[7] S. Schubert, K. Schaller, J. A. Baath, C. Hunt, K. Borch, K. Jensen, J. Brask, P. Westh, *ChemBioChem* **2023**, 24, e202200516